

5. ALGORITHM PERFORMANCE ANALYSIS

5.1 ALGORITHM BENCHMARKING

Eight algorithm variants were evaluated on SWaT and CIC-IDS datasets. To overcome extreme class imbalance, SMOTE (Chawla et al., 2002) was used before training. All supervised algorithms were used with five-fold cross validation. The latency was measured in a containerized OT simulation environment.

Algorithm	Precision	Recall	F1	Accuracy	Latency	Novel F1	CI Suitability
LSTM Autoencoder (Bidir.)	0.954	0.867	0.908	0.912	38ms	0.905	Very High
Random Forest (100 trees)	0.943	0.861	0.900	0.921	34ms	0.680	Good known partial
Federated Autoencoder (FL)	0.918	0.876	0.897	0.903	52ms	0.897	Very High privacy
GAN-Based Detector	0.885	0.871	0.878	0.881	67ms	0.845	High
Deep Autoencoder	0.932	0.817	0.871	0.874	45ms	0.871	High novel
Isolation Forest	0.873	0.740	0.801	0.800	12ms	0.793	Medium speed

Algorithm	Precision	Recall	F1	Accuracy	Latency	Novel F1	CI Suitability
Signature IDS (Snort v3)	0.953	0.869	0.909	0.906	6ms	~0.0	Known threats ONLY
One-Class SVM (RBF)	0.795	0.656	0.719	0.733	95ms	0.719	Low latency

LSTM Autoencoder achieves F1=0.908 at 38ms, which is within the 100–200ms OT threshold, with novel attack F1=0.905. Signature IDS novel F1≠0.0: cannot detect threats not in its rule database. This directly answers RQ1. Federated Autoencoder F1=0.897, just 0.011 lower than LSTM, which further proves the minimal performance loss of FL.

5.2 CONCEPT DRIFT ANALYSIS

Drift Event	Type	Pre-Drift F1	Post-Drift F1	Drop %	Recovery Days	Risk
SCADA Protocol Config Change	Sudden	0.861	0.608	29.4%	22	Critical
Sensor Replacement Flow	Sudden	0.866	0.698	19.4%	18	Critical
Firewall Rule Config Change	Sudden	0.861	0.771	10.5%	11	High
Firmware Update OT Controller	Gradual	0.871	0.794	8.8%	12	High

Drift Event	Type	Pre-Drift F1	Post-Drift F1	Drop %	Recovery Days	Risk
Firmware Update PLC v4.2	Gradual	0.859	0.799	7.0%	14	High
Network Expansion 120 IoT	Gradual	0.860	0.818	4.9%	10	Medium

SCADA Protocol Configuration Change results in 29.4% F1 degradation over 22 silent days, which is not visible on standard dashboards. The Q15 open text directly states: 'concept drift can occur due to routine changes such as sensor replacement and silently affect the performance of the AI model without raising any alarms. This is an observation made by the practitioner that supports the simulation finding.

5.3 ADVERSARIAL ATTACK ANALYSIS

Attack Type	Category	F1 Post-Attack	Evasion Rate	Composite Risk	Risk Rank
PGD Attack	ML Evasion	0.335	71.6%	0.689	1stHighest
Transfer Attack	ML Evasion	0.368	68.4%	0.643	2nd
FGSM Attack	ML Evasion	0.397	62.6%	0.601	3rd
Model Poisoning	Training Phase	0.482	55.8%	0.578	4th
False Data Injection	Data Integrity	0.549	43.8%	0.482	5th

Attack Type	Category	F1 Post-Attack	Evasion Rate	Composite Risk	Risk Rank
Replay Attack	Physical Layer	0.768	20.4%	0.268	6th
DoS on ML API	Availability	0.826	15.8%	0.228	7thLowest

The LSTM F1 drops from 0.905 to 0.335, which is a 63% drop to near random performance, when attacked by PGD. The survey result that 45.5% rate adversarial concern at 1–2 (with a mean score of 2.82/5) is a significant underestimation of this empirically-documented risk.